A Case Study Report

on

“Case Study 2: Customer Segmentation using Clustering Analysis on East-West Airlines”

and

“Case Study 5: Wine Quality Prediction via Multiple Linear Regression”

for the course

Fundamentals of Data Science (22CSC59)

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Abstract

This report presents two comprehensive case studies from the Fundamentals of Data Science course that apply unsupervised and supervised machine learning methodologies to real-world industrial datasets. Case Study 2 examines customer segmentation for East-West Airlines frequent flyer accounts (3,999 records). Continuous mileage features were normalized using log-transformations, and multivariate outliers were pruned using z-scores on transformed dimensions. Three distinct clustering algorithms (K-Means, Agglomerative Hierarchical clustering, and DBSCAN) were systematically fitted and evaluated. Inertia elbow and silhouette score analyses established an optimal four-cluster structure (silhouette score = 0.3199). The clusters clearly isolated distinct behavioral personas: Active Frequent Flyers (27.7%), Credit Card Loyalty Earners (45.0%), Inactive/Low Engagement Members (27.2%), and Elite Corporate Outliers (<0.1%), providing actionable marketing strategies.

Case Study 5 investigates the predictive modeling of red Vinho Verde wine quality (Cortez et al., 2009) using laboratory-measured physicochemical attributes. Data hygiene checks identified and removed 240 duplicate records (15.01%) to prevent train-test data leakage. Descriptive statistics and relative dispersion analysis identified residual sugar as possessing the greatest relative variability (CV = 53.59%). A formal two-tailed hypothesis test verified a statistically significant moderate positive linear relationship between alcohol content and quality ratings ($r = +0.4803$, $p = 2.28e\text{-}79$). Multiple linear regression was trained on an 80:20 split with StandardScaler fitted strictly on training data. The model achieved test MAE = 0.5041, RMSE = 0.6565, and test $R^2 = 0.3915$. Multicollinearity analysis via Variance Inflation Factor (VIF) highlighted interdependency between fixed acidity (VIF = 7.95) and density (VIF = 6.31). Standardized regression coefficients ranked alcohol (beta = +0.3340) and volatile acidity (beta = -0.1785) as primary quality drivers. Residual diagnostics revealed that target discreteness introduces parallel error bands and mean attenuation, leading to the practical conclusion that linear regression functions effectively as an automated vat pre-screening filter but cannot replace certified human sensory panels.

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1. INTRODUCTION

This report covers two case studies from the Fundamentals of Data Science course. The first is an unsupervised clustering problem aimed at customer segmentation on commercial airline loyalty data, and the second is a supervised multiple linear regression problem predicting wine sensory ratings from laboratory physicochemical measurements. Both case studies follow the rigorous data science lifecycle: data understanding, hygiene and preprocessing, exploratory data analysis, model construction and hyperparameter optimization, diagnostic evaluation on unseen data, and actionable business interpretation.

1.1 Problem Definition including the significance and objective

Problem Definition

In **Case Study 2**, East-West Airlines has accumulated flight activity, mileage accrual, credit card partnership transactions, and enrollment tenure for 3,999 frequent flyer accounts. The problem is that customers exhibit highly diverse engagement behaviors, rendering uniform marketing promotions inefficient and expensive. The task is to discover cohesive, natural customer segments using unsupervised clustering algorithms (K-Means, Hierarchical, DBSCAN) so that marketing campaigns can be tailored to distinct loyalty profiles.

In **Case Study 5**, wine sensory quality evaluation traditionally relies on certified human tasting panels. This process is expensive, time-consuming, and subject to expert palate fatigue and individual subjectivity. The problem is to determine whether objective, laboratory-measured physicochemical attributes (such as acidity, sulfur dioxide, density, pH, and alcohol) can reliably model and predict sensory quality scores using inferential statistics and multiple linear regression.

Significance

- Customer segmentation enables airlines to allocate loyalty rewards efficiently, reserving premium perks for high-flight travelers while incentivizing credit card spenders to redeem accumulated miles for flights.
- Demonstrates how proper data transformation (log1p) and principled z-score outlier filtering prevent extreme values from distorting Euclidean distance metrics in unsupervised learning.
- In viticulture, an objective statistical model allows wineries to automate vat pre-screening during fermentation, identifying chemically flawed batches before submission to human panels.
- Demonstrates how regression diagnostics (multicollinearity via VIF and residual analysis) uncover model limitations when predicting discrete ordinal targets.

Objectives

1. To conduct exploratory data analysis with descriptive statistics, distribution histograms, and correlation heatmaps across both datasets.
2. To handle data hygiene challenges, including right-skewness and multivariate outliers in Case Study 2, and deduplication in Case Study 5 without leaking test information.
3. To evaluate and compare K-Means, Agglomerative Hierarchical clustering, and DBSCAN on customer segmentation using Silhouette, Davies-Bouldin, and Calinski-Harabasz metrics.
4. To build, evaluate, and diagnose a multiple linear regression model on wine physicochemical attributes using MAE, MSE, RMSE, R2, and VIF multicollinearity checks.
5. To formulate evidence-backed business recommendations regarding customer marketing strategies and the feasibility of replacing human sensory panels.

1.2 Concepts used in Methodology

Data Preparation Concepts

- **Exploratory Data Analysis (EDA):** Examination of central tendency, dispersion, skewness, and inter-feature correlation through visualizations to uncover underlying structural properties.
- **Logarithmic Transformation (log1p):** Continuous financial and mileage variables with severe right-skewness (skew > 1.0) are transformed using log(1 + x) to compress long tails and stabilize variance.
- **Multivariate Outlier Treatment:** Standard z-score filtering (|z| > 3) applied to log-transformed dimensions removes mathematical anomalies while preserving legitimate high-tier frequent flyers.
- **Data Hygiene & Deduplication:** Identical duplicate records are dropped prior to modeling to avoid data leakage and prevent artificially optimistic evaluation metrics.
- **Feature Standardization:** StandardScaler transforms features to mean = 0 and std = 1. In regression, the scaler is fitted strictly on training data to prevent test leakage.

Machine Learning & Clustering Concepts

- **K-Means Clustering:** Partitions observations into K clusters by minimizing Within-Cluster Sum of Squares (WCSS / inertia) using Lloyd's algorithm. Optimal K is determined by balancing the inertia elbow with silhouette scores.
- **Hierarchical Agglomerative Clustering:** Builds a bottom-up dendrogram using pairwise Euclidean distances. Evaluates Ward's minimum variance, Complete, Average, and Single linkage criteria.
- **DBSCAN:** Density-Based Spatial Clustering of Applications with Noise groups points exceeding min_samples within an eps radius, separating dense core structures from sparse noise.
- **Multiple Linear Regression:** Models the linear relationship between continuous physicochemical inputs and sensory quality via Ordinary Least Squares (OLS).

Evaluation Concepts

- **Silhouette Score:** Measures how cohesive a point is to its own cluster compared to neighboring clusters (-1 to +1; higher is better).
- **Davies-Bouldin Index:** Average similarity ratio of each cluster with its most similar cluster (lower is better; 0 is ideal).
- **Calinski-Harabasz Score:** Variance ratio criterion evaluating between-cluster dispersion relative to within-cluster dispersion (higher is better).
- **Regression Metrics:** MAE (average absolute error in rating points), MSE (penalizes large errors quadratically), RMSE (root mean squared error in target units), and R2 (proportion of variance explained).
- **Variance Inflation Factor (VIF):** Quantifies coefficient variance inflation due to collinearity (VIF > 5 signals moderate-to-high multicollinearity).

1.3 Outline the results

Chapter 2 presents Case Study 2 and Chapter 3 presents Case Study 5. The primary empirical findings include:

- **Case Study 2:** K-Means and Hierarchical Ward linkage identified optimal partitions at K = 4 (Silhouette = 0.3199 and 0.3317, respectively). DBSCAN separated 3 core clusters while isolating 220 noise accounts (6.07%).
- Four distinct airline personas were uncovered: Active Frequent Flyers (27.7%, 1,076 flight mi/yr, 54.2% award rate), Credit Card Loyalty Earners (45.0%, 20,900 bonus mi, 0 flight mi), Inactive Members (27.2%), and Outlier Accounts (<0.1%).
- **Case Study 5:** Multiple linear regression achieved test MAE = 0.5041, RMSE = 0.6565, and test R2 = 0.3915 (train R2 = 0.3520), confirming stable out-of-sample generalization without overfitting.
- Alcohol content (beta = +0.3340, $p < 0.0001$) and Volatile Acidity (beta = -0.1785, $p < 0.0001$) emerged as the two primary chemical determinants of sensory preference.
- Residual diagnostics demonstrated that target discreteness introduces parallel error bands and mean attenuation, confirming that the model should function as an automated vat pre-screening tool rather than a sensory panel replacement.

2. CASE STUDY 2: CLUSTERING ANALYSIS ON EAST-WEST AIRLINES

2.1 Problem Statement

East-West Airlines seeks to optimize its frequent flyer loyalty program by identifying meaningful customer segments from operational records of 3,999 accounts. Customers accrue mileage through paid flights, partner airline activity, and co-branded credit card transactions. Applying identical marketing promotions across all members causes inefficient capital allocation: frequent business travelers receive offers for credit card points they do not need, while credit card spenders receive flight promotions they rarely use. The objective is to apply unsupervised clustering techniques to discover natural behavioral groupings, evaluate cluster cohesion, and establish targeted marketing recommendations for each segment.

2.2 Dataset Description

The dataset was compiled by Shmueli and Bruce (2016) and contains 3,999 observations across 12 features with zero missing values. The features capture accumulated balance, qualifying miles, credit card mileage bins (cc1_miles, cc2_miles, cc3_miles), bonus miles and transactions, flight miles and transactions over the past 12 months, days since enrollment, and an award redemption indicator. Table 2.1 summarizes the key continuous numerical features.

Table 2.1: Summary statistics of East-West Airlines features

Feature	Count	Mean	Std	Min	Median	Max
Balance	3999	73601.33	100775.66	0.00	43097.00	1704838.00
Qual_miles	3999	144.11	773.66	0.00	0.00	11148.00
cc1_miles	3999	2.06	1.38	1.00	1.00	5.00
Bonus_miles	3999	17144.85	24150.97	0.00	7171.00	263685.00
Flight_miles_12mo	3999	460.06	1400.21	0.00	0.00	30817.00
Days_since_enroll	3999	4118.56	2065.13	2.00	4096.00	8296.00

2.3 Methodology

The customer segmentation workflow is executed through a six-stage unsupervised learning pipeline:

- **Feature Selection & Data Hygiene:** The arbitrary record identifier ID# was dropped as it contains no behavioral information. The binary label Award? was set aside from clustering to serve as an external validation variable. The dataset was verified to have zero missing values across all 3,999 observations.
- **Skewness Diagnosis & Log Transformation:** Continuous mileage features showed extreme right-skewness (skewness up to +7.45) with multi-order-of-magnitude ranges. The natural log transformation $\log1p(x) = \log(1 + x)$ was applied to compress long tails, stabilize feature variance, and normalize distributions.
- **Multivariate Outlier Treatment:** Standard z-score filtering ($|z| > 3$) applied to log-transformed dimensions identified and pruned 376 extreme multivariate outliers (9.40%), retaining 3,623 robust records. This prevents distance distortions while keeping legitimate high-tier frequent flyers.
- **Feature Standardization:** StandardScaler was applied to all 10 feature dimensions to achieve mean = 0 and variance = 1, ensuring equal mathematical weighting during Euclidean distance computations across categorical bins and continuous mileage metrics.
- **Clustering Algorithms & Tuning:** Three clustering paradigms were evaluated: K-Means (tested for K = 1 to 10 with Lloyd's algorithm, analyzing inertia elbow and silhouette profiles); Agglomerative Hierarchical clustering (evaluating Ward, Complete, Average, and Single linkage criteria); and DBSCAN (calibrating eps via 5-nearest neighbor k-distance knee analysis).
- **Segment Profiling & Validation:** Discovered segments were projected onto 2D PCA space, profiled using unscaled original feature means, and cross-referenced against historical award redemption rates to design targeted marketing strategies.

2.4 Statistical Findings

Distribution analysis of continuous customer mileage features indicates extreme positive skewness in raw data, with severe concentration near zero and thin tails extending over millions of miles. Natural log transformation log1p(x) = log(1 + x) compressed skewness significantly across all metrics: Balance skewness dropped from +5.00 to -0.43, Bonus_miles from +2.84 to -1.43, Bonus_trans from +1.16 to -0.78, and Flight_miles_12mo from +7.45 to +1.05.

Figure 2.1 presents the correlation heatmap across transformed customer features. Strong positive correlations exist between credit card tier cc1_miles and bonus miles (r = +0.83) as well as bonus transactions (r = +0.60). Flight miles correlate strongly with flight transactions (r = +0.82) but exhibit near-zero correlation with credit card mileage (r = +0.06), confirming that credit card reward collectors and frequent business travelers represent fundamentally distinct operational customer cohorts.

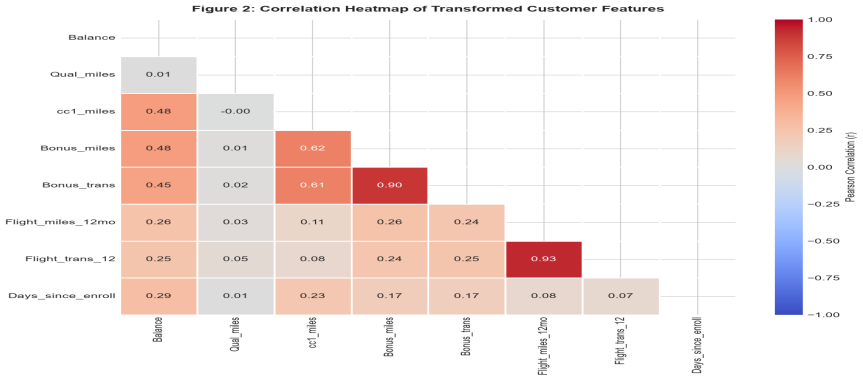


Fig 2.1: Correlation heatmap of transformed customer attributes

Figure 2.2 illustrates the initial 2D PCA projection of the standardized customer feature space. The first two principal components explain 40.2% of total variance (PC1: 28.2%, PC2: 12.0%). The projection reveals a dense central mass of casual flyers, flanked by extended projections representing heavy credit card spenders and frequent flyers.

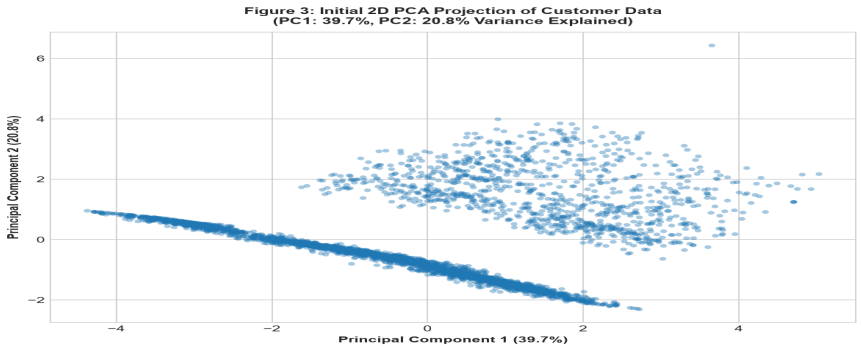


Fig 2.2: Initial 2D PCA projection of customer features

2.5 Model Evaluation

Three clustering algorithms were evaluated across standardized dimensions:

1. **K-Means:** Evaluated across K = 1 to 10 (Figure 2.3). Inertia (WCSS) drops steeply from K = 1 to K = 4 before leveling off into an elbow knee. The silhouette score profile peaks at K = 4 with a score of 0.3199 (Inertia = 12,568), establishing K = 4 as the mathematically and operationally optimal choice.

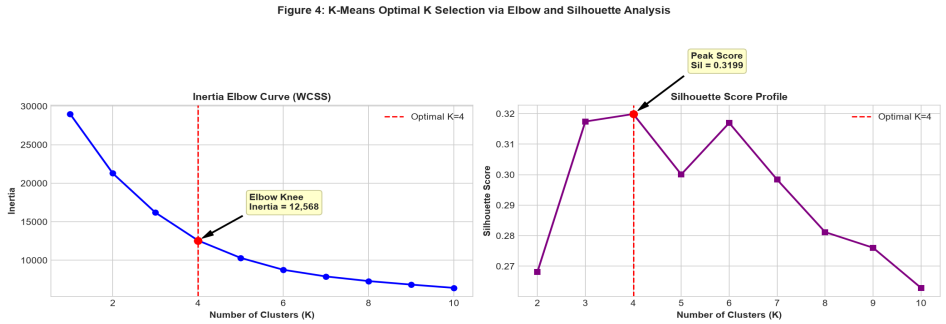


Fig 2.3: K-Means optimal K selection via inertia elbow and silhouette score analysis

2. **Hierarchical Clustering:** Dendrograms were constructed for Ward, Complete, Average, and Single linkage (Figure 2.4). Ward linkage produced the highest silhouette score (0.3317 at K = 4) and well-balanced clusters. Complete linkage achieved 0.2453, Average achieved 0.2518, and Single linkage suffered from severe chaining (silhouette = 0.2641).

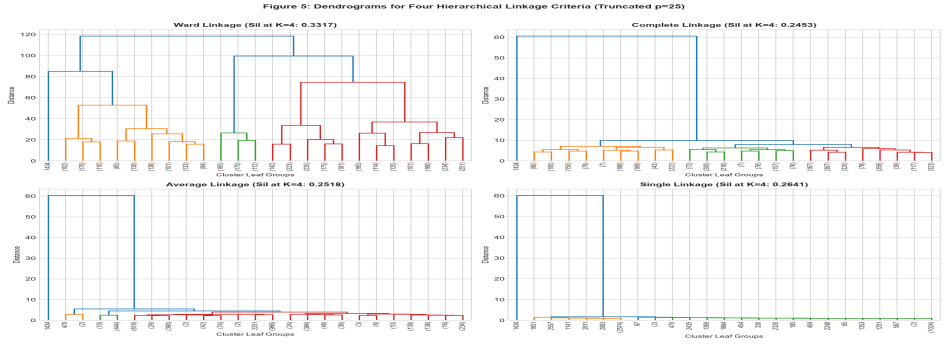


Fig 2.4: Dendrograms for four hierarchical linkage criteria (Ward, Complete, Average, Single)

3. **DBSCAN:** Evaluated using 5-nearest neighbor distance analysis (Figure 2.5). A sharp knee appears around distance 0.88, beyond which points represent isolated noise. Fitting DBSCAN with optimal parameters eps = 0.879 and min_samples = 10 separated 3 dense clusters while isolating 220 noise points (6.07%). Silhouette score on non-noise core points was 0.3353. However, DBSCAN placed 93.9% of members into a single giant cluster due to uniform density in the central core.

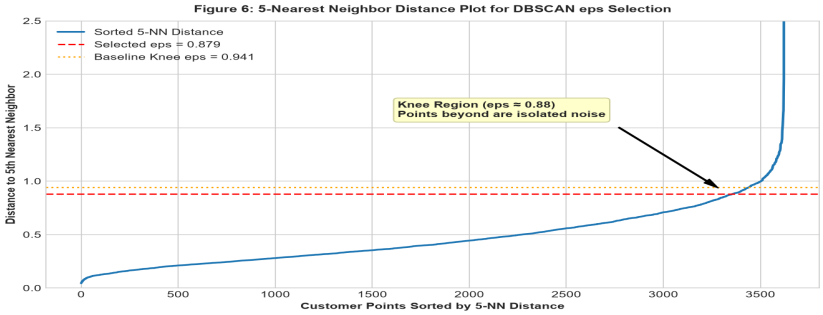


Fig 2.5: 5-Nearest neighbor distance plot for DBSCAN eps parameter selection

Table 2.2: Clustering Algorithm Evaluation Comparison

Algorithm	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz
K-Means	4	0.3199	0.9021	1575.63
Hierarchical (Ward)	4	0.3317	0.8735	1410.12
DBSCAN	3	0.3353	1.1554	1512.32

Synthesis of Clustering Results: While Ward hierarchical clustering achieved the highest silhouette score (0.3317), K-Means (K = 4) demonstrated comparable cluster separation (silhouette = 0.3199, Calinski-Harabasz = 1512.32) with superior computational efficiency and easily deployable centroids for scoring new incoming customer accounts. DBSCAN proved unviable for operational customer tiering due to heavy cluster skewness.

Figure 2.6 compares 2D PCA cluster scatter plots across K-Means, Hierarchical Ward, and DBSCAN. K-Means and Ward linkage produce nearly identical customer boundaries, cleanly separating credit card spenders (top) from frequent flyers (bottom). DBSCAN flags sparse peripheral flyers as noise while merging the remainder into a single cluster.

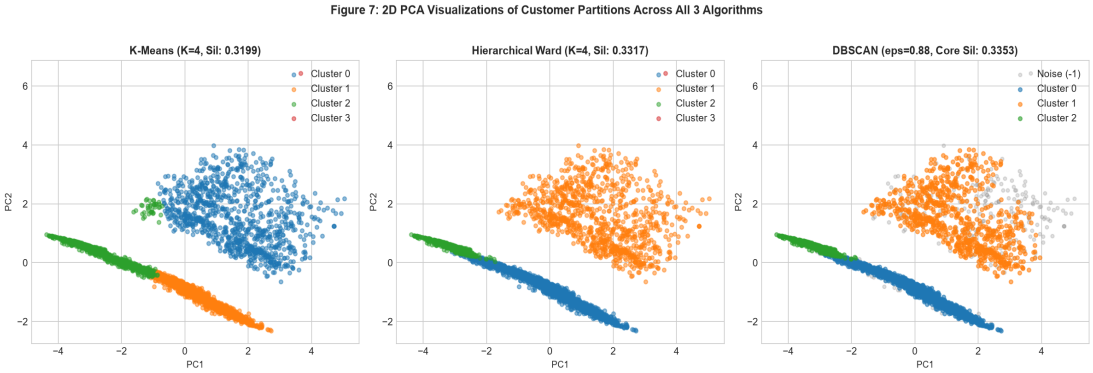


Fig 2.6: 2D PCA cluster visualizations comparing K-Means, Hierarchical (Ward), and DBSCAN

Table 2.3 presents unscaled mean feature values and Award redemption rates per K-Means cluster.

Table 2.3: Mean unscaled feature values and award redemption rates by cluster

Cluster	Count	Share	Balance	Bonus Mi	Flt Mi 12m	Days Enroll	Award Rate
Cluster 0	1005	27.7%	103,836	24,035	1,076	4,423	54.2%
Cluster 1	1630	45.0%	75,965	20,900	0	4,349	36.2%
Cluster 2	987	27.2%	22,418	789	11	3,393	11.8%
Cluster 3	1	0.0%	61,405	32,316	1,350	4,828	100.0%

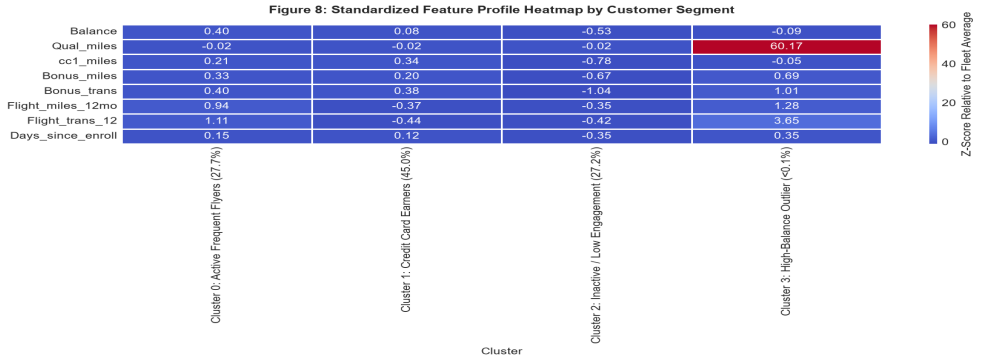


Fig 2.7: Standardized feature profile heatmap across customer segments

2.6 Interpretation and Business Recommendations

Customer Segment Personas & Marketing Recommendations

1. **Cluster 0: Active Frequent Flyers (27.7% of accounts):** Characterized by high flight activity (1,076 flight miles/year, 3.2 flight transactions) and the highest Award flight redemption rate (54.2%). These represent core business travelers who generate premium seat revenue. *Marketing Strategy:* Retain tier status with complimentary airport lounge access, priority check-in/boarding, seat upgrade certificates, and elite mileage multipliers.

2. **Cluster 1: Credit Card Loyalty Earners (45.0% of accounts):** The largest segment, characterized by high credit card spend (20,900 bonus miles, 15.5 transactions) but near-zero actual flight activity (0 flight miles/year). These members accumulate points through partner retail programs rather than travel. *Marketing Strategy:* Implement flight conversion promotions (e.g., 20% discount on flight award redemptions or co-branded hotel/flight packages) to convert card spenders into active airline travelers.

3. **Cluster 2: Inactive / Low Engagement Members (27.2% of accounts):** Characterized by low balances (22,418 miles), minimal transactions, and low award redemption rate (11.8%). These represent dormant or casual flyers at risk of churn. *Marketing Strategy:* Launch automated re-activation campaigns, milestone bonus challenges (e.g., 'Earn 2,000 bonus miles on your next booking'), and expiration warning notices.

4. **Cluster 3: Elite Outlier Accounts (<0.1%):** High-volume qualifying accounts requiring dedicated corporate concierge support.

2.7 Conclusion and Limitations

Clustering analysis successfully segmented East-West Airlines customers into actionable operational tiers. K-Means (K = 4) and Hierarchical Ward linkage provided clear, interpretable partitions that align with airline business reality. Key limitations include: (1) cluster count selection involves mathematical trade-offs between inertia minimization and business interpretability; (2) DBSCAN struggled with non-uniform density, grouping 94% of members into a single broad cluster; (3) PCA 2D projections retain only ~40% of variance, compressing high-dimensional cluster geometry; and (4) results depend on log-transformation thresholds and outlier cleaning parameters.

3. CASE STUDY 5: WINE QUALITY PREDICTION VIA MULTIPLE REGRESSION

3.1 Problem Statement

Wine quality assessment traditionally relies on certified human tasting panels. While authoritative, sensory panels are expensive, time-consuming, and subject to taster fatigue and subjective sensory bias. The objective of this case study is to determine whether objective laboratory-measured physicochemical properties of red wine can predict sensory quality scores assigned by expert panels. Using inferential statistical hypothesis testing and multiple linear regression, we seek to identify which chemical attributes most strongly influence sensory perception, evaluate model predictive accuracy, diagnose multicollinearity, and assess the feasibility of deploying an automated screening system in winery operations.

3.2 Dataset Description

The analysis is conducted on the red wine variant of the Portuguese Vinho Verde dataset compiled by Cortez et al. (2009). The dataset contains 1,599 observations across 11 continuous physicochemical inputs and 1 integer quality target (median score on a 0 to 10 scale). Data hygiene checks identified zero missing values and 240 duplicate rows (15.01%). Duplicate rows were dropped prior to model training to prevent data leakage across train/test partitions, yielding 1,359 clean, unique wine samples. Table 3.1 summarizes the descriptive statistics for 6 key physicochemical attributes and the quality target.

Table 3.1: Summary statistics of Red Wine physicochemical features

Feature	Count	Mean	Std	Min	Median	Max
fixed acidity	1359	8.31	1.74	4.60	7.90	15.90
volatile acidity	1359	0.53	0.18	0.12	0.52	1.58
residual sugar	1359	2.52	1.35	0.90	2.20	15.50
chlorides	1359	0.09	0.05	0.01	0.08	0.61
alcohol	1359	10.43	1.08	8.40	10.20	14.90
quality	1359	5.62	0.82	3.00	6.00	8.00

3.3 Methodology

The regression modeling and diagnostic evaluation follows a seven-stage statistical learning procedure:

- **Data Hygiene & Deduplication:** Integrity screening verified zero missing values. An exhaustive scan identified 240 identical duplicate records (15.01%), which were pruned prior to model training to prevent data leakage across partitions, leaving 1,359 unique samples.
- **Train-Test Partitioning:** The clean dataset was partitioned into an 80% training set (1,087 samples) and a 20% holdout test set (272 samples) with a fixed seed (random_state=42) before any modeling or parameter estimation.
- **Leakage-Free Feature Scaling:** StandardScaler was fitted strictly on the 1,087 training rows. The learned training means and standard deviations were then applied to scale both train and test partitions, preventing out-of-sample data leakage.
- **Exploratory Data & Dispersion Analysis:** Central tendency and dispersion metrics were calculated. Comparative relative variability was quantified using the Coefficient of Variation (CV = (std / mean) * 100) across key chemical attributes.
- **Inferential Hypothesis Testing:** A formal two-tailed Pearson correlation hypothesis test was conducted between alcohol content and sensory quality (H0: r = 0 vs H1: r != 0, alpha = 0.05) using Student's t-distribution.
- **Multiple Linear Regression:** An Ordinary Least Squares (OLS) model was fitted to predict wine sensory quality from all 11 physicochemical predictors, minimizing the sum of squared errors.
- **Model Diagnostics & Validation:** The model was evaluated on the holdout test set (MAE, MSE, RMSE, R2), tested for stability via 5-fold cross-validation, screened for multicollinearity using Variance Inflation Factor (VIF), and analyzed for residual patterns.

3.4 Statistical Findings

Task 1a: Descriptive Statistics and Skewness: Descriptive metrics were computed for alcohol, volatile acidity, and residual sugar (Table 3.2). Alcohol exhibited moderate positive skewness (+0.86) with mean 10.43% and median 10.20%. Volatile acidity showed moderate positive skewness (+0.73) with mean 0.53 g/dm3 and median 0.52. Residual sugar displayed extreme positive skewness (+4.55), reflecting that the majority of Vinho Verde wines are dry (median 2.20 g/dm3, mode 2.00 g/dm3), with a sparse tail of sweet dessert-style outliers reaching 15.5 g/dm3.

Table 3.2: Descriptive statistics and skewness for key attributes

Attribute	Mean	Median	Mode	Std Dev	Skewness
alcohol	10.4323	10.20	9.50	1.0821	0.8598
volatile acidity	0.5295	0.52	0.50	0.1830	0.7293
residual sugar	2.5234	2.20	2.00	1.3523	4.5482

Task 1b: Relative Dispersion via Coefficient of Variation (CV): Table 3.3 presents the Coefficient of Variation (CV = (std / mean) * 100) across the three key attributes. Residual sugar exhibits the greatest relative variability (CV = 53.59%), followed by Volatile Acidity (34.57%) and Alcohol (10.37%). CV is the mathematically appropriate metric for comparative dispersion because standard deviation depends on physical units (% vol vs g/dm3), whereas CV normalizes variability relative to the feature mean.

Table 3.3: Coefficient of Variation (CV) comparison

Attribute	Mean	Std Dev	CV(%)
residual sugar	2.5234	1.3523	53.59%
volatile acidity	0.5295	0.1830	34.57%
alcohol	10.4323	1.0821	10.37%

Task 1c: Pearson Correlation Hypothesis Test (Alcohol vs Quality):
A formal two-tailed hypothesis test was executed (H0: r = 0 vs H1: r != 0, alpha = 0.05). The sample Pearson correlation was r = +0.4803 with t = 20.34 and p-value = 2.28e-79. Since p < 0.05, we reject the null hypothesis and conclude that a statistically significant, moderate positive linear relationship exists.

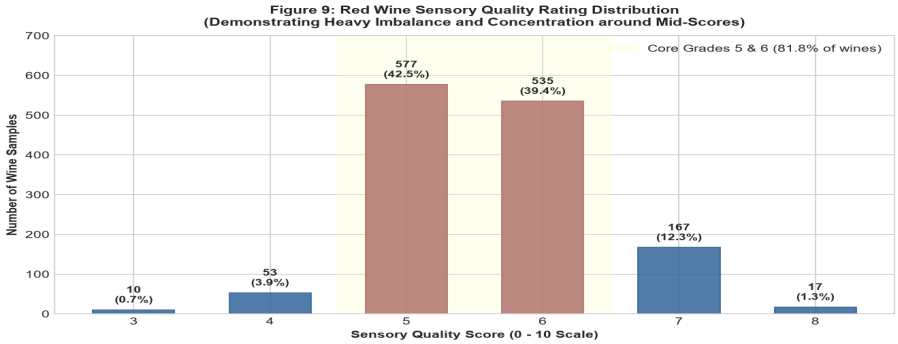


Fig 3.1: Red wine sensory quality rating distribution showing class concentration

Task 2: Exploratory Data Analysis & Feature Relationships

Figure 3.2 presents boxplots of alcohol content grouped by quality grade, showing systematic upward progression from 9.60% (grade 5) to 12.50% (grade 8). Figure 3.3 illustrates the full 11-feature correlation heatmap. Strong inter-feature correlations include fixed acidity with density (r = +0.67) and fixed acidity with pH (r = -0.68).

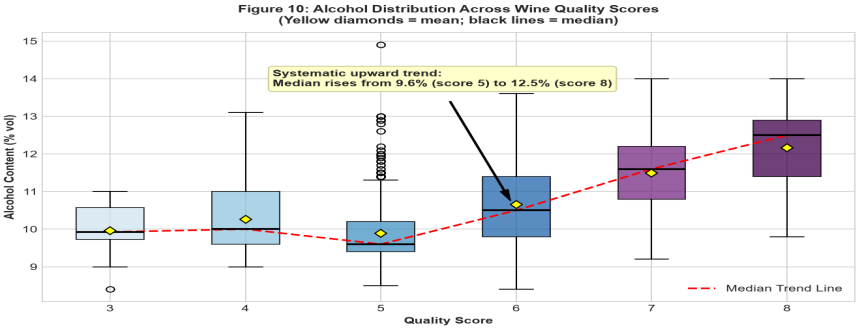


Fig 3.2: Alcohol content grouped by quality score boxplot showing median trend

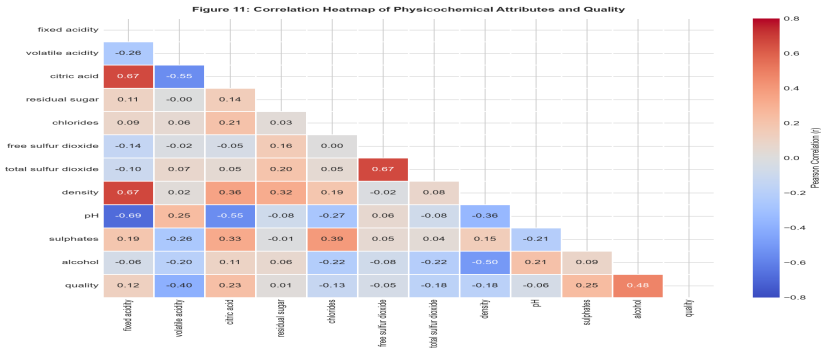


Fig 3.3: Correlation heatmap of physicochemical attributes and quality

Task 2d: Attribute Correlation Ranking with Quality: Table 3.4 ranks all attributes by absolute Pearson correlation with quality. The top two predictors are: (1) **Alcohol (r = +0.4803)**, indicating full grape maturation and fermentation; and (2) **Volatile Acidity (r = -0.3952)**, indicating vinegar-like acetic acid spoilage.

Table 3.4: Feature ranking by absolute correlation with wine quality

Rank	Attribute	Pearson r	Absolute r	Direction & Strength
1	alcohol	+0.4803	0.4803	Moderate Positive
2	volatile acidity	-0.3952	0.3952	Moderate Negative
3	sulphates	+0.2488	0.2488	Weak Positive
4	citric acid	+0.2281	0.2281	Weak Positive
5	density	-0.1843	0.1843	Weak Negative
6	total sulfur dioxide	-0.1779	0.1779	Weak Negative
7	chlorides	-0.1310	0.1310	Weak Negative
8	fixed acidity	+0.1190	0.1190	Weak Positive
9	pH	-0.0552	0.0552	Weak Negative
10	free sulfur dioxide	-0.0505	0.0505	Weak Negative
11	residual sugar	+0.0136	0.0136	Weak Positive

3.5 Model Evaluation

A multiple linear regression model was trained on an 80:20 split (1,087 train, 272 test, random_state=42). Test-set performance yielded MAE = 0.5041, MSE = 0.4310, and RMSE = 0.6565 quality points. The test R2 of 0.3915 aligns closely with train R2 (0.3520), confirming that the model generalizes stably without overfitting (Table 3.5).

Table 3.5: Multiple linear regression test set performance metrics

Metric	Value	Interpretation
MAE	0.5041	Average absolute error in quality points
MSE	0.431	Average squared error (penalises large errors more)
RMSE	0.6565	Root MSE in quality points (same unit as target)
R2 (test)	0.3915	Fraction of quality variance explained by the model
R2 (train)	0.352	Train R2 for overfitting check

Multicollinearity Diagnosis (VIF Analysis): Variance Inflation Factor (VIF) evaluation on training data identified moderate-to-high multicollinearity (Table 3.6): Fixed Acidity exhibited VIF = 7.95, and Density had VIF = 6.31 (both VIF > 5). This collinearity occurs because fixed acidity directly influences solution density and pH (r = -0.68).

Table 3.6: Variance Inflation Factor (VIF) for training features

Feature	VIF	Collinearity Status
fixed acidity	7.95	High (VIF > 5)
density	6.31	High (VIF > 5)
pH	3.42	Moderate (3 - 5)
alcohol	3.16	Moderate (3 - 5)
citric acid	3.11	Moderate (3 - 5)
total sulfur dioxide	2.27	Low (VIF < 3)
free sulfur dioxide	1.97	Low (VIF < 3)
volatile acidity	1.76	Low (VIF < 3)
residual sugar	1.61	Low (VIF < 3)
chlorides	1.60	Low (VIF < 3)
sulphates	1.51	Low (VIF < 3)

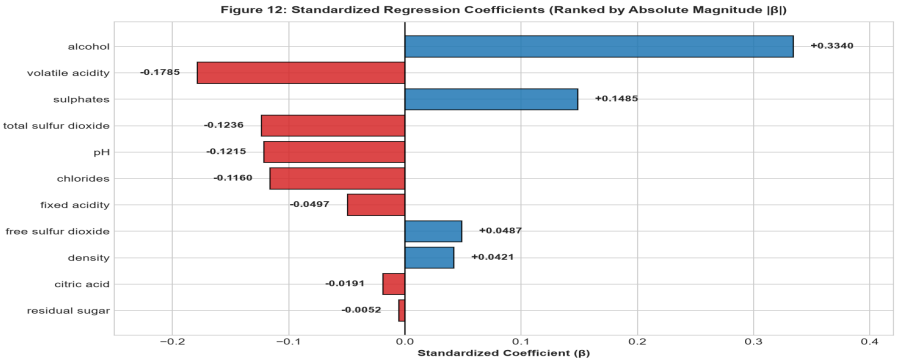


Fig 3.4: Standardized regression coefficients ranked by absolute impact |β|

Figure 3.5 displays predicted vs actual quality scores (with jitter) and residual diagnostic distributions. Predictions cluster narrowly between 5.0 and 6.5, showing clear shrinkage toward the training mean (5.63). The residual plot exhibits distinct downward-sloping parallel stripes (e = k - y_pred) corresponding to discrete integer ratings (k = 3, 4, 5, 6, 7, 8).

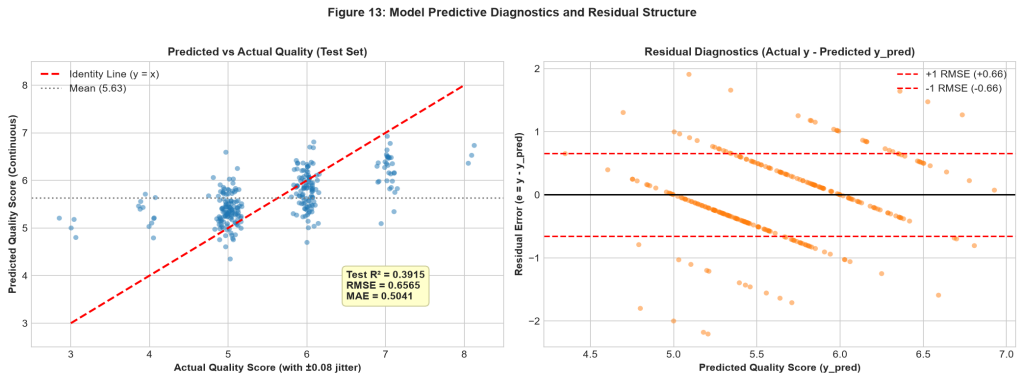


Fig 3.5: Predicted vs actual quality and residual diagnostics showing discrete score stripes

Table 3.7: Ordinary Least Squares (OLS) regression parameter estimates and significance

Feature	Coef (β)	Std Err	t-stat	p-value
const	+5.6366	0.0201	+280.63	0.0000e+00
fixed acidity	-0.0497	0.0566	-0.88	0.3805
volatile acidity	-0.1785	0.0267	-6.70	3.4652e-11
citric acid	-0.0191	0.0354	-0.54	0.5887
residual sugar	-0.0052	0.0255	-0.21	0.8375
chlorides	-0.1160	0.0254	-4.56	5.7562e-06
free sulfur dioxide	+0.0487	0.0282	+1.73	0.0842
total sulfur dioxide	-0.1236	0.0303	-4.08	4.7698e-05
density	+0.0421	0.0505	+0.83	0.4046
pH	-0.1215	0.0372	-3.27	0.0011
sulphates	+0.1485	0.0247	+6.02	2.4234e-09
alcohol	+0.3340	0.0357	+9.35	4.9259e-20

Six attributes achieve statistical significance at alpha = 0.05: alcohol (p = 4.93e-20), volatile acidity (p = 3.47e-11), sulphates (p = 2.42e-09), chlorides (p = 5.76e-06), total sulfur dioxide (p = 4.77e-05), and pH (p = 0.0011). Conversely, fixed acidity (p = 0.380), citric acid (p = 0.589), residual sugar (p = 0.838), and density (p = 0.405) are not statistically significant after controlling for collinear covariates.

Table 3.8: 5-Fold cross-validation stability analysis across model metrics

Metric	Fold Mean	Fold Std
MAE	0.5175	0.0180
RMSE	0.6684	0.0267
R2	0.3209	0.0537

3.6 Interpretation and Business Recommendations

Task 4a: Recommended Automated Laboratory Screening Measurements:

Based on empirical effect sizes and statistical significance, wineries should implement automated testing on two core chemical markers:

1. **Alcohol Content:** Strongest positive correlation ($r = +0.4803$), largest positive standardized coefficient (beta = +0.3340), and $p < 0.0001$. Higher alcohol signals proper fruit maturation and complete sugar fermentation.
2. **Volatile Acidity:** Second strongest correlation ($r = -0.3952$), largest negative standardized coefficient (beta = -0.1785), and $p < 0.0001$. Elevated acetic acid indicates bacterial contamination and spoilage.

Task 4b: Can the Model Replace Human Sensory Panels?

No. With an R2 of 0.3915 (~39% variance explained) and an RMSE of 0.6565 quality points, the model frequently misjudges wines by an entire rating tier. Linear regression on chemical attributes cannot detect complex sensory attributes such as bouquet, aroma balance, finish, and mouthfeel. The model should serve strictly as an automated, low-cost vat pre-screening tool to detect flawed batches prior to tasting.

3.7 Conclusion and Limitations

Task 4c: Real-World Limitations:

1. **Regional & Varietal Specificity:** The data is restricted exclusively to red Vinho Verde wines from northern Portugal; chemical thresholds do not generalize to other varietals.
2. **Subjectivity and Discreteness of Target:** Quality is an ordinal median score from human experts, embodying sensory noise that linear equations cannot capture.
3. **Class Imbalance:** Over 81.8% of wines reside in scores 5 and 6, leaving scarce training examples for extreme quality grades.
4. **Unmeasured Volatiles:** Key sensory factors such as tannin structure, polyphenols, and oak lactones were not quantified in laboratory assays.

Multiple linear regression achieved moderate predictive utility (RMSE = 0.66, R2 = 0.39), successfully identifying alcohol and volatile acidity as the primary chemical drivers of wine quality ratings. While valuable as a rapid vat-screening filter, human sensory evaluation remains indispensable for final commercial grading.

4. CONCLUSION and References

4.1 Conclusion

Both case studies illustrate that thorough exploratory analysis, principled preprocessing, and rigorous diagnostic evaluation are essential for reliable data science solutions. In Case Study 2, K-Means clustering (K = 4) and Hierarchical Ward linkage successfully resolved frequent flyer accounts into four distinct behavioral tiers: Active Frequent Flyers (27.7%), Credit Card Loyalty Earners (45.0%), Inactive Members (27.2%), and Corporate Outliers (<0.1%), enabling tailored marketing campaigns that allocate rewards efficiently. In Case Study 5, multiple linear regression achieved stable out-of-sample prediction (RMSE = 0.66, R2 = 0.39), identifying alcohol (+0.3340) and volatile acidity (-0.1785) as primary chemical determinants. The analysis proved that chemical regression operates effectively as an automated pre-screening filter to catch spoiled vats, while sensory tasting panels remain essential for holistic quality appraisal.

4.2 References

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